

Atharva Kulkarni

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SUMMARY

Software Engineer experienced in designing, building, and operating AI-driven, scalable distributed systems using Java, Python, and cloud-native patterns. Delivered high-performance APIs and data processing pipelines on AWS, automated testing, CI/CD workflows, and observability practices. Passionate about writing clean, maintainable code using object-oriented principles and leveraging AI-assisted development tools.

SKILLS

- Programming Languages:** Python, Java, JavaScript, TypeScript, SQL, C++, HTML, CSS
- Backend:** Node.js, Express, Java Spring Boot, Django, Flask, REST APIs, gRPC, GraphQL, WebSockets
- Databases and Cloud:** Microsoft SQL Server, MySQL, PostgreSQL, MongoDB, Firebase, AWS EC2, Lambda, S3, SQS, Docker, Kubernetes, CI/CD, GitHub Actions
- AI & Tools:** React.js, GitHub Copilot, Cursor, Claude, Git, Postman, Jest, JUnit, Jira, Agile, DevOps, Kafka

EDUCATION

- Syracuse University** | Master of Science, Computer Science | (GPA: 3.67/4.0) **Aug 2023 - May 2025**
- University of Mumbai** | Bachelor of Eng., Computer Engineering | (GPA: 9.41/10) **Aug 2019 - Jun 2023**

WORK EXPERIENCE

Syracuse University **Syracuse (NY), USA (Remote)**
Software Engineer **Jun 2025 - Present**

- Architected an AI-driven resume screening platform using Python and spaCy NER, building data ingestion pipelines that processed large volumes of student data and reduced manual review time by 75%
- Designed scalable PostgreSQL schemas and Node.js REST APIs using object-oriented principles, achieving sub-200ms response times with observability logging
- Built automated data processing workflows to normalize and match 2,000+ job postings to resume profiles, leveraging AI-assisted development tools to accelerate feature delivery

Clutch Delivery LLC **Syracuse (NY), USA**
Software Engineering Intern **May 2024 - Aug 2024**

- Eliminated backend bottlenecks by restructuring Node.js APIs, optimizing MongoDB queries with indexing, and offloading static assets to Amazon S3, reducing latency from 450ms to 130ms
- Implemented DevOps workflows by containerizing with Docker and building CI/CD pipelines with GitHub Actions, integrating automated testing, static checks, and rollback capabilities
- Designed and operated event-driven, cloud-native architecture on AWS (SQS, EC2 auto-scaling, Lambda, CloudWatch) with observability instrumentation, processing 500+ concurrent orders
- Increased daily active users by 40% by redesigning and rebuilding customer, driver, and admin platforms in React.js and TypeScript with modern component-based architecture

TCR Innovation **Mumbai, India**
Full-Stack Web Developer Intern **Jan 2022 - Jul 2022**

- Doubled backend throughput by optimizing Java Spring Boot microservices, tuning PostgreSQL queries and indexes, and adding caching on high-traffic routes for faster responses
- Added JUnit for unit and integration tests for Spring Boot services, increasing backend test coverage to 80% and reducing production defects and regressions after release
- Enabled 1,000+ monthly reservations by building backend services for a travel booking platform, including Stripe payments and reservation persistence, with a React-based frontend for user interaction

The Sparks Foundation **Mumbai, India**
Web Development Intern **Jul 2021 - Dec 2021**

- Improved banking platform performance by 55% by optimizing Java Spring Boot APIs and redesigning PostgreSQL schemas with indexing, supporting 1,200+ concurrent transactions
- Architected a secure authentication system with JWT-based sessions, timeout policies, and multi-layer validation for operations, ensuring reliability and compliance

PROJECTS

Syracuse University Shuttle Tracker | Java, Spring Boot, MS SQL Server, React.js, MVC

- Designed and deployed a real-time shuttle tracking system using WebSocket connections serving 120+ daily active students and processing 2500+ monthly requests, improving transportation efficiency
- Cut shuttle wait times by 30% by creating a dynamic drop-off algorithm and validating reliability through unit testing the API with JUnit, ensuring smoother scheduling

Skin Cancer Classification using CNN | Python, TensorFlow, Node.js, React.js, OpenAI

- Built a full-stack classification platform using a Convolutional Neural Network (CNN) with the HAM10000 dataset, achieving 85% prediction accuracy with OpenAI API to generate diagnostic insights
- Deployed the model via a Express.js backend and React.js interface using TensorFlow.js for real-time image analysis, ensuring secure on-device image processing across 500+ test cases